The Impact Of Socioeconomic Indicators On GDP Per Capita Across Countries

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Department of Mathematics
University of Ruhuna, Matara
Supervisor: Dr. K.D.Prasangika

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Introduction

Background of the study

- GDP per capita is the key indicator of the nation's economic health.
- This study explores how socioeconomic indicators influence GDP per capita across countries.
- Our study examined sample of 154 countries to uncover key relationships.

Introduction

Significance of the study

- This study helps to identify which factors have the most impact on GDP per capita.

Importance of the study

- Provides evidence based insights for improving economic planning and development of a country.
- The findings emphasize the vital link between social welfare and economic output.

Research Question

How does key socioeconomic indicators influence GDP per capita across different countries?



Aim and Objectives

Aim

- To identify the impact of key socioeconomic indicators on GDP per capita.

Objectives

- To determine the importance and impact of each socioeconomic factor on economic performance across countries.
- To develop multiple linear regression model to quantify the relationship between these indicators and GDP per capita.
- To identify and select the most effective key socioeconomic features for our analysis.

Literature Review

Author & Year	Key Findings
Sari & Cahyadin (2021)	• Life expectancy increases with GDP per capita • Unexpected link between child mortality and GDP
Batrancea et al. (2021)	• Exports have strongest positive impact on GDP per capita • Imports and domestic savings show positive effects • FDI impact was less clear
Bhide & Khanolkar (2020)	• Human net migration, Literacy rate Death rate , Number of phones per 1,000 people show positive effects on GDP per capita.

Research Gap

- Most of these analyses have been limited to specific countries, regions, or smaller sets of variables.
- This study aims to fill this gap by analyzing a wider range of socioeconomic variables across varied sample of nations to provide a more understanding of their influence on GDP per capita.

Method

Research design

- This study has a quantitative research design.
- Multiple linear regression was used to investigate the relationship between the dependent variable and the independent variables.
- This method empowers researchers to uncover patterns, relationships, and trends within the data.

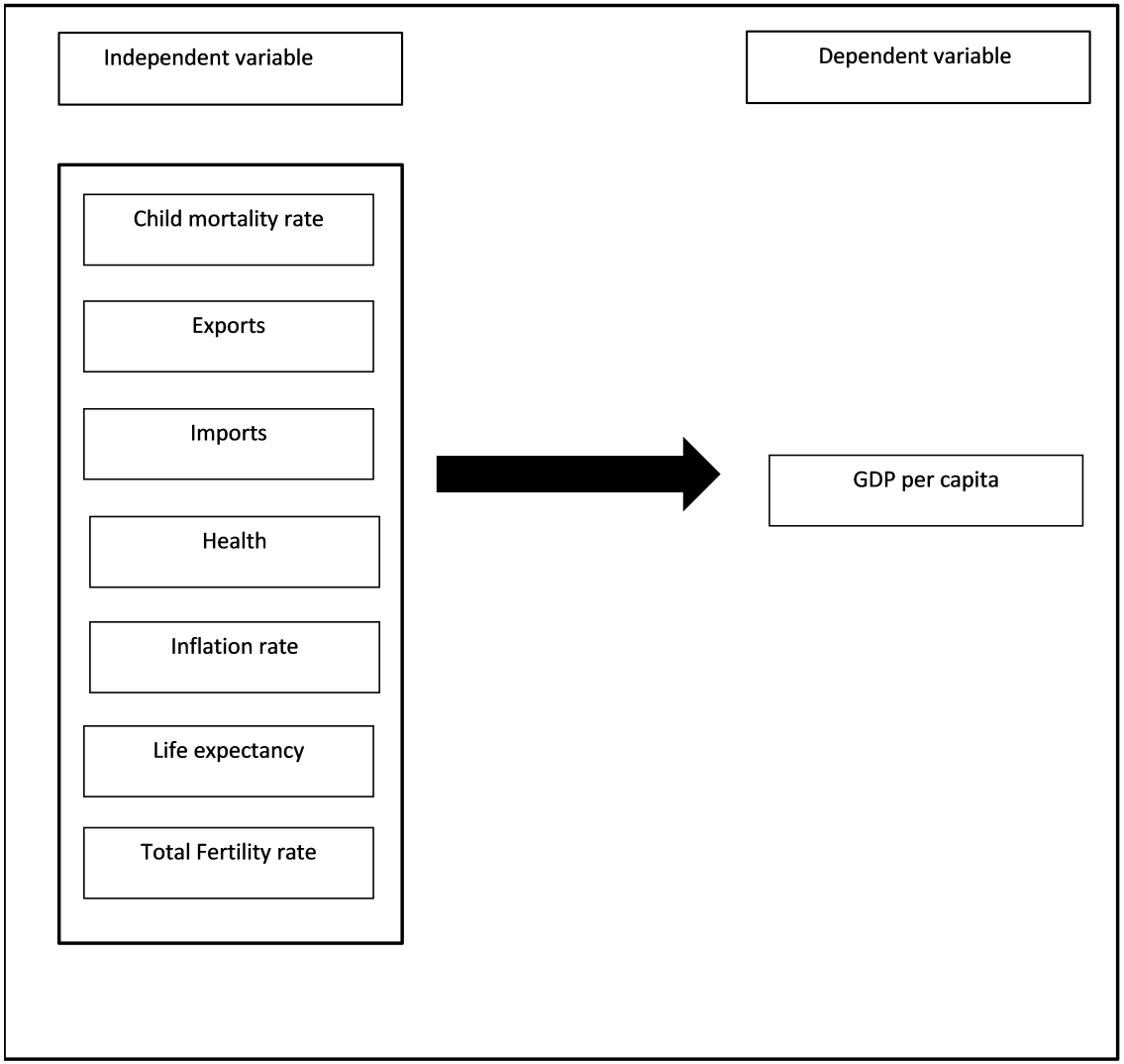
The Multiple linear regression model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon \tag{1}$$

where:

- Y is the dependent variable,
- $X_1, X_2, \ldots, X_p$ are the independent variables,
- β_0 is the intercept,
- $\beta_1, \beta_2, \ldots, \beta_p$ are the coefficients of the independent variables,
- ε is the error term.

The conceptual model



Variable Description

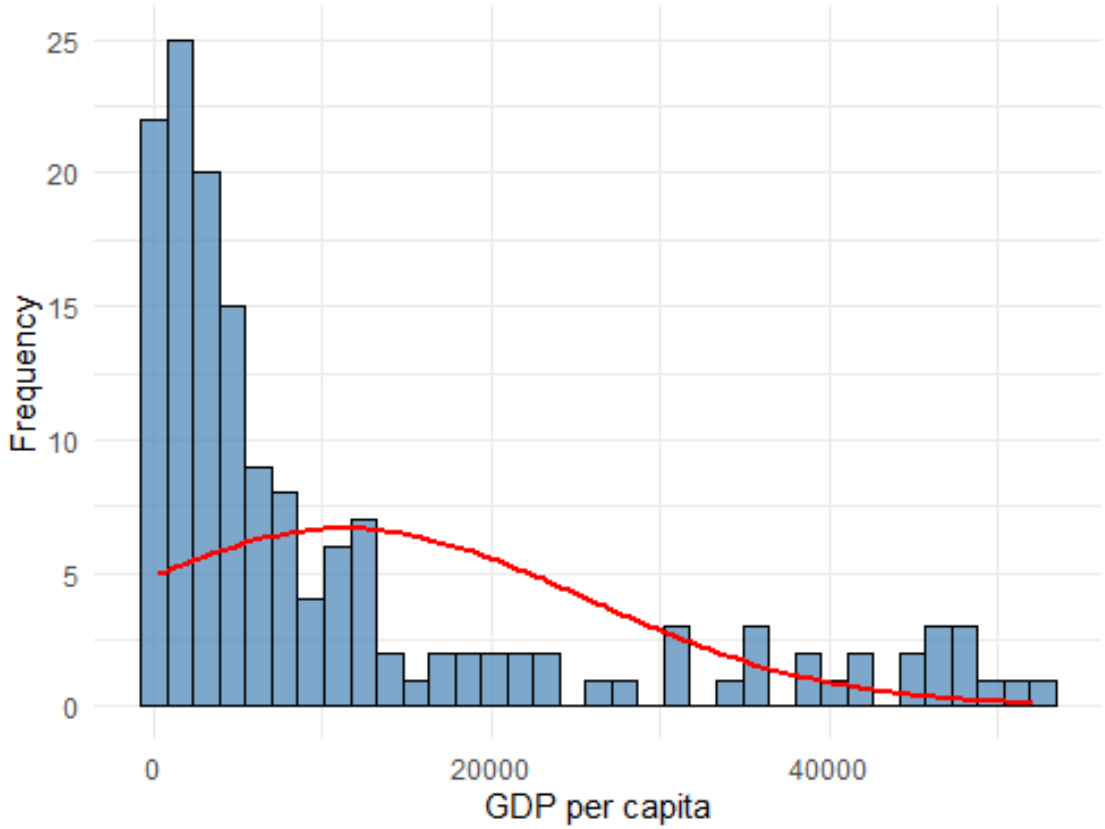
Independent variables	**Dependent variables**
Child mortality rate	GDP per capita
Imports	
Exports	
Inflation rate	
Income	
Life expectancy	
Health	
Total fertility rate	

Data set preparation

- The child mortality rate and total fertility rate were transformed by taking their inverse values.
- It helped to linearize their relationship with the dependent variable.
- We check whether this data set has null value by using R software.
- Additionally, during the data processing stage, some extreme outliers were identified and removed.

Analyzing dependent variable

Variable	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
gdpp	11334	1151	14289	231	1340	4670	13200	52100



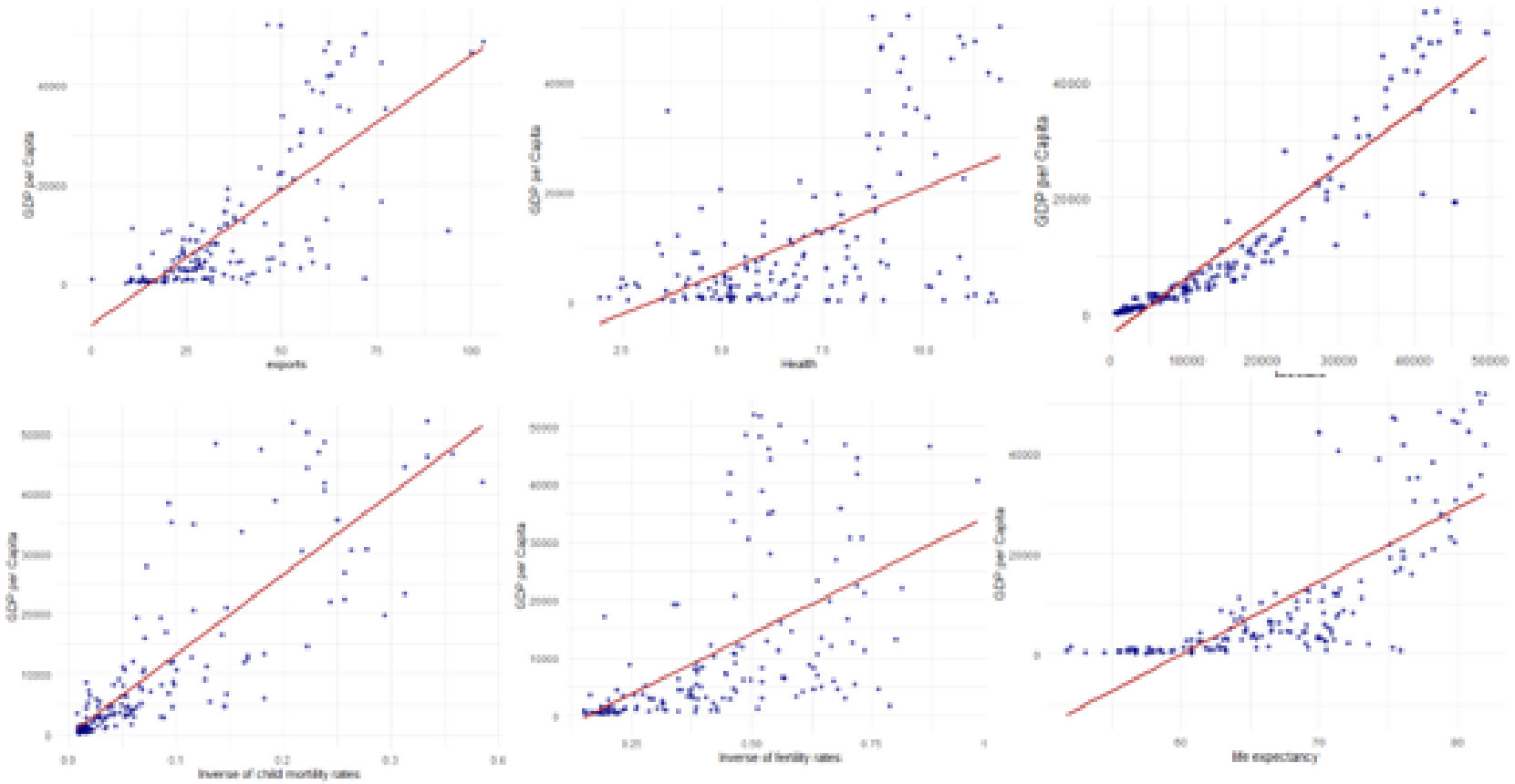
Diagnostic tests for model

To ensure our model is valid, we first checked for several key assumptions.

- **Linearity**
  A linear relationship between the dependent and independent variables.
- **Normality of Residuals**
  The residuals of the model are normally distributed.
- **Homoscedasticity**
  The variance of the residuals is constant.
- **No Multicollinearity**
  The independent variables are not highly correlated with each other.

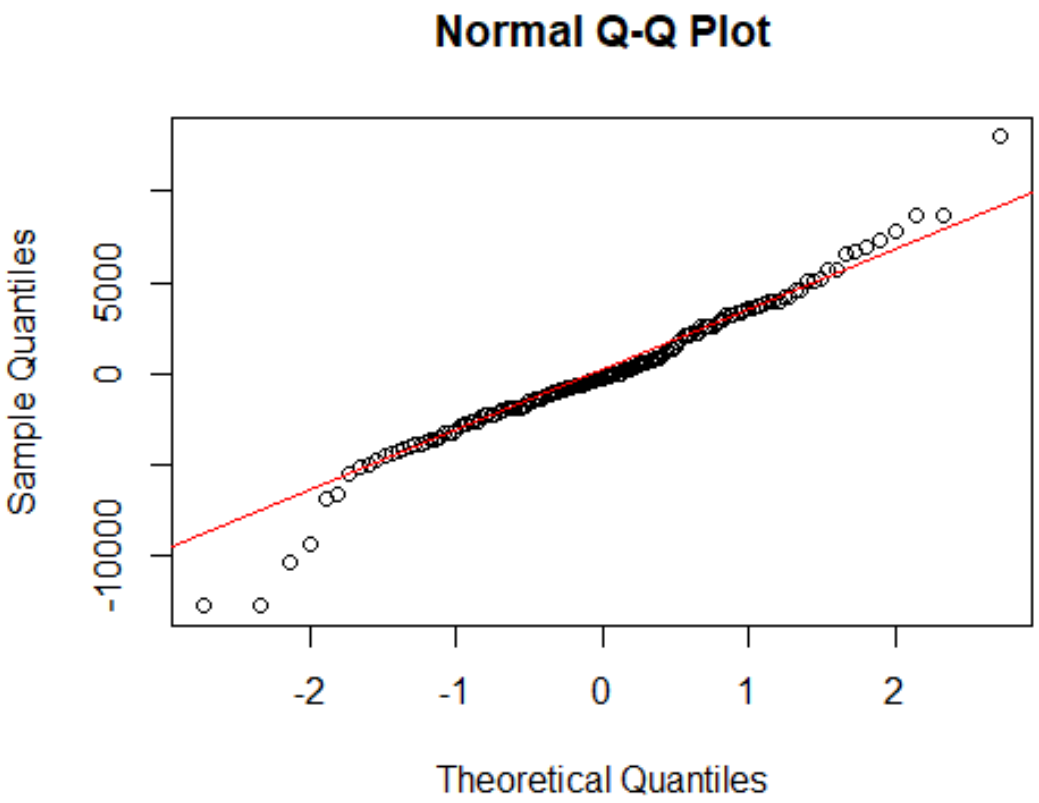
Results

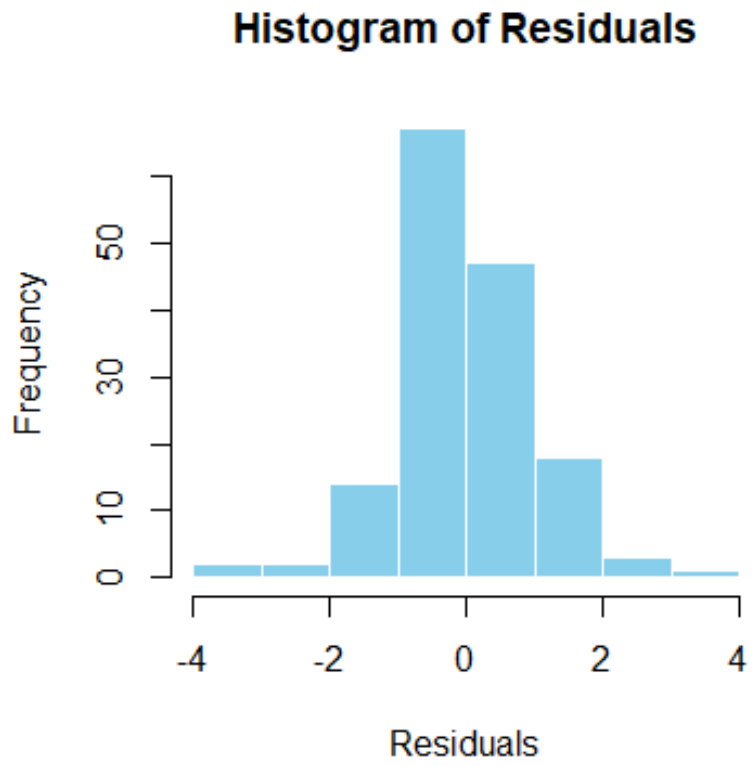
1 Linearity



There is a linear relationship between the dependent and independent variables.

2 Normality of Residuals





Considerable Q-Q plot points over the line and the histogram of residuals It shows residuals follow a normal distribution.

3 Multicolinearity

- Value greater than 10 is generally considered indicative of problematic multicollinearity.
- No variable exceeds 10, so overall, the model does not exhibit severe multicollinearity.

Variable	VIF
exports	3.016059
health	1.453825
imports	1.406635
income	5.358015
inflation	1.489495
life_exp	3.290196
inverse child_motility	4.377896
inverse total fer	2.154702

Hypothesis 1

$$H_0 : \beta(i) = 0, \quad i = 1, 2, 3, 4, 5, 6, 7$$

There is no linear relationship between dependent variable and independent variables.

$$H_1 : \beta(i) \neq 0, \quad i = 1, 2, 3, 4, 5, 6, 7$$

There is a linear relationship between dependent variable and independent variables.

Results

Construct multiple linear regression model considering all Independent variables

Model	Beta	Estimate	$Pr(>\|t\|)$
(Intercept)	$\hat{\beta}_0$	260.296	0.879113
exports	$\hat{\beta}_1$	106.139	5.97e-07
health	$\hat{\beta}_2$	697.718	$< 2 \times 10^{-16}$
imports	$\hat{\beta}_3$	-88.735	1.10e-07
income	$\hat{\beta}_4$	0.663	0.000233
inflation	$\hat{\beta}_5$	-222.276	0.000498
inv_child_mort	$\hat{\beta}_6$	41409.816	1.32×10^{-5}
inv_total_fer	$\hat{\beta}_7$	-13000.211	1.70e-05
life_expt	$\hat{\beta}_8$	0	0.56406

Fitting the Best Regression Model

- We have used Backward elimination method for building an accurate model.
- A low p-value (typically less than 0.05) indicates that the parameter has a significant impact on the dependent variable.
- Life˙expectancy was removed in our final regression Model.
- Coefficient of determination of the reduced model shows 92.7%

```
Model   => gdpp ~ exports + health + imports + income + inflation + life_exp+
           inv_child_mort + inv_total_fer
R2      => 0.927

Removed => life_exp

Model   => gdpp ~ exports + health + imports + income + inflation +
           inv_child_mort + inv_total_fer
R2      => 0.92703
```

Our Reduced Model

$$\begin{aligned}\text{GDP per capita} = 260.296 &+ 106.139 \cdot \text{Exports} \\ &+ 697.718 \cdot \text{Health} \\ &- 88.735 \cdot \text{Imports} \\ &+ 0.663 \cdot \text{Income} \\ &- 227.357 \cdot \text{Inflation} \\ &+ 41409.816 \cdot \text{Inverse Child Mortality} \\ &- 13000.211 \cdot \text{Inverse total fertility rate}\end{aligned}$$

Hypothesis 2

- **Null hypothesis (H0):**
  Reduced model is suitable

- **Alternative hypothesis (Ha):**
  Full model is suitable

Results

- First considered the F partial.

$$F = \frac{(SSE_{reduced} - SSE_{full})/(p - m)}{SSE_{full}/(n - p - 1)} = 0.3343$$

- Then got the F table value.

$$F(1 - 0.05, 1, 145) = 3.906392$$

$$\text{The F partial} < \text{F table}$$

$$0.3343 < 3.906392$$

- Failed to reject null hypothesis at 0.05 significance level.
- The reduced model is suitable.

Discussion

- We began descriptive analysis, visualizing relationships through plots and summary statistics.
- During the data preprocessing stage, some extreme outliers were removed to reduce their influence on the regression results.
- We did correlation analysis, estimate models, model summaries, ANOVA tables by using R software.
- To get the best fitted model, We used Backward elimination method.
- Life expectancy was removed since it was not significant.
- Assumptions mostly satisfied except Homoscedasticity ,Reduced model was still accurate.

Conclusion

- The final model explains 92.7% of the variance in GDP per capita indicating this is a very strong fit.
- Income and exports showed the strongest positive association with GDP per capita.
- Inverse Child mortality, Inverse total fertility rate and Health expenditure were positively linked with GDP per capita.
- Also imports and inflation were negatively associated with the GDP per capita.
- These findings highlight the necessity of policies that increase income and exports, improve healthcare to reduce child mortality and carefully manage import dependency and inflation.

References

- Batrancea, L., Rathnaswamy, M. M., Batrancea, I.(2021). A panel data analysis of economic growth determinants in 34 African countries.
- Bhide, N., Khanolkar, S. (2020). A comprehensive study on the factors impacting the GDP (per capita) of major economies around the globe using regression analysis. International Journal of Computer Applications, 175(10).
- Sari, V. K., Cahyadin, M. (2020). The impact of health on per capita GDP in Indonesia.

Thank You!

Q & A

